

Open discussion for the reading:

Needle point example. how an unstandard way of presenting data can help us better understand uncertainty but also how uncomfortable people are with simplifying complicated data.

Interesting how performative visualizations can be used to present data

A student realized that the amount of people that dislike the jitter demonstrates that it really helped capture uncertainty, and that we are typically uncomfortable with uncertainty. Also questions if this is a good representation if the overall reaction is negative.

A student also questioned where the line is drawn for ethics and technology, with how big companies' actions don't match up to the message they conveyed.

- Lead us to discuss about how LLMs might be the last straw that might have helped us recognize the importance of recognizing ethics or critical thinking when approaching stem topics
- Another student also recognized that LLM helped us bring humanities back into the light. And it helped us recognize how intrusive LLM are and detrimental reliance on it is, that it's even present when texting or writing emails.
- Discussion on how these reliance might have limited our development of critical thinking outside of the scope of technical domains.

Discussion: can a visualization ever be neutral

The student's annotation, that even in the data collection step there might not have been neutrality, so it questions if any visualization could be neutral.

- Student voice that it's vague what neutral even is. Namely looking at the school shooting data set that is targeted at evoking emotions, so if the intent of the data was to evoke some emotion it doesn't really feel neutral anymore.

Follow up can we make a visualization that is not evoking any emotion:

- Student: Can't really think of a visualization that doesn't evoke any emotion. Because even if we take a bland random graph it can evoke certain emotions in people so maybe it can never be neutral of emotions but maybe we could get close.

Talked about contrast between the two graphs like how the visualization difference between the two graphs about gun deaths might have different levels of neutrality or less emotion.

- Students voice how the topic of the data makes it inherently not neutral.
- Hard to do neutrality with the gun death examples because the human component makes it so we already go in with some emotions. Extend to how even with more bland topics like plants or geology might connect to other human related topics. So the underlying relation to humans always makes it hard to feel neutral about data.

Student: the reason why visualization can't be neutral is that we are missing so much information such as confounding effects. So we can never get a full story from the data.

Student: visualization can't really be neutral because we always go into building them with the intention of presenting some message.

Anecdote about visualization from Dr. T. Lab science graphs that use classic themes, that even if they can be cleaner and objective, the small pieces that we include like grid color all serve to present some message.

Discussion question: If there's no neutrality in visualization what should we be asked to provide alongside the data visualizations

Student suggestion:

- showing who is affected by the data
- Descriptions like how we have in the museum, such as backgrounds to the data scientist or background into the data. However it might be hard to enforce this because not everyone might be transparent about their agenda specifically in politics.
 - Example how we might want to employ this in relation to the unemployment graph
 - Student suggestion: maybe presenting the newspaper they were made for and the people who made it to give context on what the graph might be arguing for.
 - Student: Since they are meant to tell a story maybe we don't need it to be neutral but we can look at both sides of the story and get a fuller picture based on them.

Student question: if there could be dishonesty in presenting data and if we can add it to the code of ethics from last week.

- Makes us question in what spaces can we enforce these rules or changes. Maybe it can't be enforced in politics but it can be enforced in scientific journals. Maybe we can enforce this to make scientists justify their choices. But this prompts the question of what it might look like if we wanted to make this change.
 - Students mentioned the positionality statement, and maybe presented their own background and quick explanation about choices in analysis. Overall something short might be more expressive.
 - Student: good papers typically have justifications for their approach, especially since not explaining them might undermine the validity of the paper. And that maybe the author's background might not have been included because the target audience might not have any interest in this information.
 - A positional statement might need to show how your lived experience might have shaped the person you are and it might be helpful in seeing how this impacted your research, and offer a deeper intellectual connection between you and the work that you're doing than online research.

Question from student: if it's more neutral to include these positionality statements. Like if including this information might damage the audience's perspective.

- There's always going to be some bias from readers, however the paper is going to prove the same thing no matter what. So transparency might help make work more objective but at the same time it could be possible that it makes your work more dismissive.
- Student: this also relates to the responsibilities of readers, that they are actually reading with the intent of learning something instead of just confirmation bias.
- Dr T mentions how this might actually humanize the work, and help readers connect with the research better.
- We all bring in lived experience and unconscious bias that might shape the research in some way. But it's definitely in the reader's discussion to decide if they want to read the paper. But it's not the author's responsibility to persuade the readers to avoid bias.
- Dr T the lack of positionality in stem paper might be an example of how they express objectivity. But it's important to recognize ourselves as subjective pieces.
- Mention discussion with classmates who have had to get a lot of text books for varying opinion and reading. It kind of shows how stem is just presented as much more of an objective field and we've just gotten used to it.
- Dr T mentions that there is indeed space for us to learn stats in a different setting.
- Students mentioned that learning with a more human related sample might also be important in grasping concepts.
- Dr T mentions how even the plot title can evoke varying reactions, because it's basically an instruction for how to read the plot. That even when trying to say the plot title in a neutral way it doesn't mean that what you're showing is not neutral. It's important to recognize that there's a lot of places where objectivity can show up and it's up to us to recognize where they could show up for us.
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Got into how text books might be neutral but even in text book examples we don't really see neutrality.

Has anyone been asked to make a visualization that is emotional

Student: It's interesting to see her girlfriend make plots in the mustang news and see the process of making graphs in a context where it is meant to express emotion.

Talked about data fest: the winning team showing how long and how many times they needed to have other symptoms to find their diagnosis.

Also talked about eviction cases, giving a human component to data by adding stories behind each point.